PADMAVATHI DHANASEKAR

AI & DATA SCIENCE STUDENT • INTERNSHIP CANDIDATE

Chennai, Tamil Nadu, India | padmavathi.tech.official@gmail.com
GitHub: github.com/padmavathi-Tech08 | LinkedIn: linkedin.com/in/padmavathitech
LeetCode: leetcode.com/u/Padmavathi_tech/

PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence and Data Science student with practical experience developing Python-based applications and data-driven projects. Skilled in Python, SQL, Pandas, NumPy, exploratory data analysis, and machine-learning fundamentals, with a strong foundation in DSA, OOP, and Operating Systems. Seeking an internship to apply technical and problem-solving skills in Data Science, AI/ML, or software development.

EDUCATION

B.Tech – Artificial Intelligence and Data Science | Adhi College of Engineering and Technology

2025–2029 • CGPA: 8.87

TECHNICAL SKILLS

Programming: Python, C
Data: SQL, Pandas, NumPy
Data Visualization: Matplotlib, Seaborn, Plotly
Data Science: Data Cleaning, EDA, Feature Engineering, Statistical Analysis
Machine Learning: ML Fundamentals, Classification, Regression, Model Evaluation
Tools: Jupyter Notebook, Git, GitHub
Core Concepts: DSA, OOP, Operating Systems

PROJECTS

Smart City Traffic Optimizer | Python

- Developed a route-optimization project for finding efficient paths in a dynamic city road network.
- Implemented graph-based algorithms including Dijkstra's algorithm, BFS, and A*.
- Added traffic updates and road closures using structured JSON data.
- Integrated network visualization to represent routes and road conditions.

SafeRoute – Smart Safety Navigation System | HTML, CSS, JavaScript, Python, Flask, SQLite

- Developed a safety-focused navigation web application that recommends routes using safety scores and route factors.
- Implemented Safest, Balanced, and Fastest route options with explainable safety scoring.
- Incorporated lighting, population, reported safety, and route reliability into route evaluation.
- Implemented SOS alerts, community safety reports, journey saving, safety tips, and dashboard features.
- Built a responsive frontend connected to a Python Flask backend and SQLite database through REST APIs.

AI-Based Smart Logistics & Accessibility Platform | AI/ML

- Designed an intelligent logistics platform concept addressing transportation and accessibility challenges in the North Eastern Region.
- Proposed route accessibility analysis considering difficult terrain, weather conditions, and road blockages.
- Designed AI-driven approaches for logistics visibility and intelligent route recommendations.
- Focused on improving transportation planning for essential goods and remote-area accessibility.

AREAS OF INTEREST

Artificial Intelligence • Machine Learning • Data Science • Problem Solving

LANGUAGES

English • Tamil